

Temporally Constrained, Provenance-Verified Evidence Retrieval for Indian Legal Research

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Abstract—AI-assisted legal research fails in ways that fluent output conceals: a cited authority may not exist, may lack the attributed passage, may post-date the matter, or may never have been retrieved. A 2026 Supreme Court of India ruling overturned tribunal findings built on exactly these defects, making citation verifiability a functional requirement. This paper presents an evidence pipeline for historical Indian Supreme Court judgments in which every displayed citation is temporally eligible, provenance-bound, and mechanically verifiable against the persisted corpus. The design is extractive: material propositions are shown verbatim with locators, and outcome prediction is kept separate so a classification score can never stand in for evidence quality. We report three findings on a combined 37-case reference-evidence population (frozen 30-case base plus additive 7-case extension, analysed in strata). First, naive identifier matching between ILDC and an eCourts-derived judgment archive is unsafe: of 5,391 syntactic matches only 11 survived content alignment, motivating a reusable gate. Second, pre-ranking temporal eligibility with salient-term queries and a self-match guard raised Recall@5 from 5/30 to 12/30 and Recall@100 from 12/30 to 15/30 on the frozen base (12/37 and 20/37 combined). Third, all 185 displayed citations passed grounding, provenance, and temporal verification, with zero unsupported claims across the 37 cases, while 17 expected authorities remained absent at $k=100$: verification and retrieval must be measured on separate denominators. Outcome baselines on 1,503 ILDC cases and inference-time evidence augmentation are reported with their own populations and limits. A blinded 14-case, four-LLM-rater exploratory comparison favoured structured presentation 56/56, reported as presentation evidence only.

Index Terms—legal information retrieval, explainable AI, provenance verification, temporal filtering, retrieval-augmented systems, Indian legal NLP, BM25, InLegalBERT

I. INTRODUCTION

Legal research operates in a domain where fluency is not evidence. A generated response can be linguistically convincing while citing a judgment that does not exist, attributing a passage to a real judgment that does not contain it, or relying on law that was unavailable at the time the matter arose.

Empirical profiling of legal hallucination in large language models has established that these failures are frequent rather than exceptional [1], and related work has probed how far explicit legal standards can be communicated to language models at all [2].

The problem is no longer confined to the literature. In *Pooja Ramesh Singh v. Jammu and Kashmir Bank Ltd. & Anr.*, 2026 INSC 668, the Supreme Court of India set aside tribunal decisions after identifying nonexistent authorities, incorrect citations, and fabricated passages apparently produced through AI-assisted research [3]. A legal research tool must therefore produce evidence a reviewer can locate, inspect, and independently verify.

This paper treats that requirement as the primary design constraint over historical Indian Supreme Court judgments: case facts are retrieved with stable locators, duplicates excluded, only selected verbatim passages rendered, each citation verified against the persisted corpus and retrieval run, and outcome prediction kept in a separate field that never alters the evidence-bound conclusion.

The central empirical observation is a dissociation that a single aggregate score would erase. Under the final configuration, all 185 displayed citations across the combined 37 cases cleared grounding, provenance, duplicate, and temporal checks, with no unsupported claim among the 37 cases; at the same time, 17 of the 37 independently verified expected authorities never appeared among the top 100 candidates returned by the retriever. Correctness of what is shown and completeness of what is found are therefore separate properties. Reporting only a combined “citation correctness” figure would hide exactly the gap a reader needs in order to decide how much to trust the output.

Contributions. This work makes four contributions, each bounded to the evaluated setting.

- 1) **A quantified cross-corpus identity failure and its**

remedy. Converting eCourts identifiers of the form YYYY INSC N to ILDC-style YYYY_N produced 5,391 syntactic candidates, of which only 11 passed content alignment (Section IV-C): identifier equality is unsafe for this dataset pair. We supply a reusable content-alignment gate based on title/party identity plus direct phrase overlap.

- 2) **Pre-ranking temporal eligibility.** We implement historical availability as a predicate inside the BM25 candidate relation rather than as a post-hoc filter, so that ineligible judgments cannot consume retrieval depth (Section VI-D). On the held-out evidence set this change moved Recall@5 from 5/30 to 12/30 without regressing any prior success.
- 3) **A four-state citation protocol with separate denominators.** Fabricated, real-but-unretrieved, policy-violating, and valid-but-unexpected citations (Section VII) — four states that a single correctness label collapses.
- 4) **An evaluation that keeps recovery and integrity apart.** Authority recovery, citation integrity, outcome prediction, and explanation presentation are reported on their own populations, documenting the resulting dissociation rather than resolving it into a headline number (Section VIII).

Scope. Claims are deliberately narrow: no legal correctness for retrieved authorities, no autonomous legal advisor, no claim that evidence retrieval improves outcome prediction. Ours is a researcher-facing prototype on four non-pooled populations: 1,503 outcome cases, 30 frozen plus 7 additive source-verified evidence cases in strata, and paired explanation examples (seven formative author-reviewed pairs, superseded by a 14-case exploratory LLM comparison).

II. RELATED WORK

A. Indian legal NLP and judgment prediction

ILDC established Indian Supreme Court judgment prediction and explanation as a benchmark task [4], supplying the fixed splits and labels used here. InLegalBERT [5] motivates our neural baseline; practitioner-evaluated summarisation [6] shows length and presentation are first-order concerns. None of these lines addresses cross-corpus identity or per-citation provenance, which is where this work sits.

B. Legal retrieval, benchmarks, and temporal constraints

COLIEE evaluates case-law and statute retrieval and entailment [7]; LegalBench measures LLM legal reasoning [8]; CaseHOLD tests holding selection and pre-training value [9]. Each uses one fixed corpus and none requires a candidate to pre-date the query matter. TaxFlow applies hybrid RAG with validity filtering to Indian tax-law question answering [10]; CaseFacts frames U.S. legal claim verification with Supported/Refuted/Overruled labels [11], showing validity evolves; LexTime benchmarks temporal event ordering [12]. Retrieval-augmented generation established retrieval as an explicit non-parametric evidence source [13]; BM25 supplies

the sparse ranking basis [14] and Tesseract the OCR engine for corpus repair [15].

C. Positioning

The boundary against the closest work is as follows. We claim neither the first legal RAG system, nor the first Indian legal model, nor the first temporal legal benchmark: the contribution is the reproducible combination of content-alignment gating, conservative pre-ranking eligibility, exact retrieval-run verification, and separate recovery-versus-validity reporting on a source-first key. Jurisdictions, tasks, corpora, and denominators differ, so no external system is used as a numerical baseline. Per-work boundaries: ILDC/CJPE [4] supplies the prediction setting and splits (we add a separate provenance-bearing corpus without reproducing its architecture); InLegalBERT [5] motivates the neural baseline (we correct truncation via chunk-and-pool without assuming it must win); TaxFlow [10] shares legal-retrieval concerns (statutory validity; we address precedent identity, passage provenance, and a verified key instead); CaseFacts [11] shows validity evolves with noisy open retrieval (we verify identity, eligibility, and provenance from case facts); LexTime [12] treats event ordering, whereas we enforce availability through metadata before ranking.

III. PROBLEM FORMULATION

Let a historical query judgment have identifier q , query year Y_q , and frozen facts-only text F_q . Let each candidate evidence passage p belong to a source judgment with a stable source identifier, an exact decision date, citation metadata, page and character boundaries, and stored passage text.

The temporal policy admits p only when the decision year of its source is strictly earlier than Y_q . A same-year source is treated as ambiguous, because ILDC supplies no exact query date; a source with missing or unparseable date metadata is excluded. Both rules trade recall for the guarantee that no displayed authority can post-date the matter.

The evidence task decomposes into three separable questions.

- **Recovery.** Does the top-100 candidate set contain the independently verified expected authority, and is that authority among the five displayed sources?
- **Integrity.** Does each displayed citation reproduce a real retrieved corpus passage with exact provenance, avoid the query judgment and its aligned duplicates, and satisfy the temporal rule?
- **Presentation.** Given identical evidence and citations, does a structured Issue–Authority–Evidence–Conclusion–Uncertainty format make the material easier to inspect than unstructured prose?

Outcome prediction is a secondary evaluation capability, not a fourth research question. It is reported on its own population and never pooled with the evidence measures, so that a classification score cannot substitute for evidence quality and a traceable citation cannot be mistaken for complete authority

TABLE I
OPERATIONAL DEFINITIONS USED THROUGHOUT THE FROZEN
EVALUATION.

Term	Implemented definition
Facts-only input	Text retained before the earlier of a recognised dispositive cue and a 60% character cap, sentence-aligned where possible; inputs below 10% retention or 100 words are excluded
Temporal existence	The source has a parseable exact decision date in corpus metadata
Temporal applicability	For query year Y , an authority is eligible only if its decision year is $< Y$; same-year, later-year, and missing-date sources are excluded
Temporal effectiveness	The eligibility predicate is applied inside the BM25 candidate relation before ranking and <code>LIMIT 100</code> , so ineligible sources cannot consume returned depth
Recall@100	Share of answer-key cases whose predefined authority occurs anywhere in the returned top-100 candidates
Recall@5	Share of answer-key cases whose predefined authority is among the five selected and displayed sources
Provenance validity	A displayed item reproduces the stored source ID, citation, decision date, court, PDF/page/character locator, exact passage, and retrieval-run membership
Citation groundedness	Every material displayed proposition is verbatim text of a supplied evidence passage and links to that evidence item
Authority consistency	A verified displayed source matches the answer-key authority by stable source ID, normalised citation, or normalised title plus exact decision date

recovery. Table I fixes the operational definitions and pass/fail criteria.

A. Research questions

There are exactly three research questions. **RQ1:** Does grounding an Indian legal AI workflow in retrieved legal evidence improve legal-research reliability and relevant-evidence retrieval compared with a facts-only legal-language baseline? The controlled comparison is E2 versus E3; E1 is a traditional baseline reported for context. **RQ2:** Does provenance-constrained evidence selection and citation verification reduce unsupported or unverifiable legal claims in the final output? **RQ3:** Can the proposed structured explanation format improve human-verifiable transparency without materially degrading prediction or retrieval performance?

For RQ3, the implemented comparison holds the underlying evidence and citations constant and compares structured with unstructured presentation. It evaluates presentation transparency only; it does not independently establish a measurable prediction or retrieval-performance degradation or non-degradation effect.

IV. CORPUS CONSTRUCTION AND AUDITING

A. Purpose-split corpus design

We use two corpora for different experimental purposes.

ILDC Single supplies fixed case-level splits and binary outcome labels: 7,593 judgments (5,082 training, 994 validation, 1,517 test). After a shared sufficiency rule excluded 14 test records, the held-out prediction population contains 1,503 cases.

The evidence corpus is an eCourts-derived collection of Indian Supreme Court judgments covering 1950–2020, supplying the metadata absent from ILDC: citations, exact decision dates, court and case identifiers, source paths, and page and character locators. Of 39,069 English PDF instances, 39,066 accepted PDFs yielded 2,343,435 labelled chunks, of which 2,036,981 unique chunks were loaded into a PostgreSQL provenance store and a SQLite FTS5 BM25 index.

This split follows the information actually available in each source. ILDC supports fixed-split outcome prediction but exposes only a year in the case identifier and lacks the citation and passage provenance that verification requires. The eCourts collection supports dated, traceable retrieval but is not used as a substitute outcome-label benchmark. Cross-corpus links are drawn only after the alignment checks described below.

B. Source-quality audit and targeted OCR repair

A full audit found the stored text valid UTF-8, but 15 PDF instances (representing 14 source IDs) had missing or badly corrupted embedded text, concentrated in image-backed 1980s/1990s Reports plus one mojibake-affected 2018 file. Fixed checks covered absent embedded text, control characters, mojibake markers, visible-ASCII ratio, and English-token ratio.

Only flagged instances were re-rendered and processed with English Tesseract OCR at 250DPI [15]. Twelve passed the same post-OCR quality gate and were restored; three single-page PDFs remained below threshold and were transparently excluded rather than forced into the index. Raw PDFs were preserved unchanged, exclusions were recorded, and the provenance store and BM25 index were rebuilt from the accepted corpus.

C. Cross-corpus identifier alignment

Linking ILDC query cases to eCourts sources requires establishing that two records describe the same judgment. The obvious route — converting an eCourts identifier of the form `YYYY INSC N` to the ILDC-like string `YYYY_N` — is unsafe, and quantifiably so.

That conversion produced 5,391 syntactic candidates. **Only 11 passed content alignment; 5,380 were identifier-namespace collisions**, concentrated in legacy material where similar numeric suffixes routinely denote entirely different judgments. Trusting identifier equality here would silently merge unrelated cases, corrupting both deduplication and leakage control.

The corrected pipeline treats identifier equality as candidate generation only. A reusable gate evaluates title or party

identity together with direct six-token phrase overlap before any mapping may drive deduplication or retrieval exclusion. Combining syntactic and title/party candidates yielded 8,927 deduplicated pairs, of which 1,304 were accepted and 7,623 rejected. Retrieval additionally applies a full-document self-match safeguard, so that an unmapped copy of the query judgment can be removed without suppressing an earlier authority merely quoted at length by the later judgment.

We regard this audit as independently useful. It demonstrates that mechanically similar ILDC and eCourts identifiers do not establish judgment identity, and it supplies a content-based alternative reusable across answer-key validation, development probes, leakage checks, and retrieval-time exclusion.

D. Source-verified authority answer key

The authority evaluation uses 30 ILDC fixed-test cases. Test membership was confirmed before external verification. For each accepted case the query judgment was inspected through a primary source or accepted eCourts mirror, one earlier authority was recorded *independently of system retrieval*, and that authority was reconciled to the corpus by stable source ID, normalised citation, or normalised title plus exact date for parallel reporter forms.

The identifier-collision discovery triggered a read-only audit of the then-current mappings: 20 passed, nine resolved sources failed direct content alignment, and one source was unresolved. One case was relinked to a content-aligned source and the other nine flagged records were replaced by new fixed-test cases, yielding a final audit of 30/30 direct-content passes.

The sample is not era-balanced: five query cases are from the 1970s, 13 from the 1980s, nine from the 1990s, two from the 2000s, and one from the 2010s. The 1980s account for 43.3%. This concentration is a consequence of the source and alignment gates rather than deliberate temporal sampling, and we treat it as a limitation (Section X).

A later additive extension verified seven further fixed-test cases (1990_234, 1990_256, 1990_324, 1991_136, 1991_87, 1992_286, 1993_90) under the same gates, without modifying the original 30-case key (listed in Section VIII). Three further candidates were held back as needing review and are not part of any evaluation. The combined 37-case population is therefore a later expanded analysis, not the original frozen base; all results below retain Base-30, Extension-7, and Combined-37 strata.

V. SYSTEM DESIGN

The system comprises four auditable layers. E1 is the traditional TF-IDF plus logistic-regression outcome baseline; E2 is the facts-only InLegalBERT outcome baseline. E3 is the retrieval/evidence-grounded branch, and E4 is the verification, provenance, and structured-explanation bundle applied to that branch. E2 is not a retrieval baseline. Retrieval quality is evaluated against the predefined authority/evidence key; outcome prediction is secondary.

1. Corpus and alignment layer. ILDC splits supply prediction cases and labels. Cleaned eCourts chunks retain stable

source, date, citation, court, PDF, page, and character provenance. A content-aligned crosswalk plus a runtime content check control target-case leakage.

2. Outcome branch. A shared facts-only extractor feeds sparse linear baseline (E1) and InLegalBERT chunk-and-pool model (E2). These baselines produce binary predictions and never retrieve external evidence.

3. Evidence branch (E3). Facts-only text is converted to salient terms, matched through BM25 under pre-ranking temporal eligibility, and narrowed to five source-diverse passages rendered in a fixed evidence-linked structure.

4. Verification and reporting layer (E4). Each displayed record is verified against the corpus and the recorded retrieval run; duplicate and temporal rules are enforced; verified sources are compared with the source-first answer key. Versioned artifacts preserve metrics, failures, and configuration provenance.

The evidence renderer is extractive: material legal text is displayed verbatim, and its evidence-bound conclusion uses fixed, non-inferential wording tied to evidence identifiers. The renderer cannot retrieve, rerank, invent an authority, paraphrase a material proposition, or infer an outcome. Outcome prediction is produced separately by the prediction branch.

VI. METHOD

A. Shared facts-only input

E1 and E2 share the extraction configuration `ildc-predecision-facts-v1`. The extractor retains text preceding the earlier of a recognised dispositive cue and 60% of the document, moving the boundary to a preceding sentence end when sufficient text remains. Cases below 10% retention or 100 words are excluded. Neither prediction path uses retrieval. Model and threshold selection uses validation data only; the held-out test split is evaluated once, after selection.

B. E1: sparse linear baseline

E1 uses lowercased TF-IDF unigrams and bigrams with sublinear term frequency, minimum document frequency two, L_2 normalisation, and at most 100,000 features. Logistic-regression values $C \in \{0.1, 1.0, 10.0\}$ are compared by validation accuracy, with smaller C breaking ties. The selected $C = 10.0$ pipeline is refitted once on eligible training and validation cases and evaluated once on test data. Seed 202605.

C. E2: InLegalBERT chunk-and-pool

E2 fine-tunes `law-ai/InLegalBERT` (revision `b5ecfed8`) with a new two-label head. It supersedes an earlier 256-token prefix run that truncated 99.20% of eligible test inputs — a discarded result retained for traceability. Each facts-only document is instead represented by overlapping 512-token windows with 50-token overlap. Training uses three epochs, seed 202607, learning rate 2×10^{-5} , weight decay 0.01, warm-up ratio 0.1, and gradient accumulation.

Mean pooling of window logits before softmax is the primary rule, and checkpoint selection uses validation document-level mean-logit accuracy only. Majority vote is reported as a

secondary comparison, not a post-hoc selection rule. All 1,503 eligible test documents and 9,576 test windows are covered.

D. E3: temporally constrained retrieval

The query builder `tfidf-segment-salient-terms-v1` segments the full facts-only input, removes procedural-report boilerplate, scores terms deterministically, retains section and article cues, and emits at most 32 unique terms. These query a SQLite FTS5 BM25 index backed by PostgreSQL provenance.

The frozen configuration is `week11-bm25-salient-terms-preranked-temporal-v3`: only judgments with decision year strictly earlier than the query year enter the candidate relation *before* BM25 ordering and the top-100 cutoff, so the full depth is spent on eligible material rather than discarded ineligible depth.

Candidate sources are checked against the alignment-gated crosswalk and a direct self-match rule requiring at least 100 shared six-token occurrences and 80% unique candidate-source phrase coverage. A non-learned selector then chooses up to five passages in BM25 order, with at most one passage per source.

E. E4: citation and provenance verification

E4 receives the E3 answer, retrieval-run identifier, query identity and year, and persisted corpus records. Each displayed item must exist in the corpus, reproduce exact passage and provenance, belong to the recorded run, avoid the query and duplicates, and pre-date the query year; any failure rejects the citation.

Authority consistency is evaluated separately, after verification. A fully valid citation may differ from the single expected authority without being labelled substantively irrelevant — a distinction we return to in Section VIII-D.

F. Inference-time evidence augmentation

Outcome prediction was subsequently added to E3 and E4 by reusing the frozen E2 checkpoint without fine-tuning. A shared input builder concatenates the frozen facts extract and the selected verbatim evidence in selection order, applies the same 512-token windows with 50-token overlap, pools window logits by their mean, and selects the class by `argmax`. Gold labels, scores, dates, provenance, and boilerplate are excluded from model input. Seed 202607; configuration `e3e4-evidence-augmented-inlegalbert-checkpoint6318-v1`.

We note the methodological caveat plainly: the checkpoint was trained on facts-only inputs, so applying it to facts plus retrieved passages is an inference-time distribution shift, not evidence-aware fine-tuning. These predictions are therefore reported descriptively — on the frozen 30-case base and, in the RQ1 extension, on the combined 37 cases — using the frozen E2 checkpoint at inference time, and are not directly comparable with the 1,503-case baselines, which form a separate population with a different denominator.

G. Bundled-intervention caveat

E4 jointly checks corpus existence, exact passage/provenance identity, run membership, duplicate status, and temporal eligibility over shared final retrieval and evidence. Aggregate reliability cannot be attributed to any one component; the supported claim concerns the complete bundle.

VII. CITATION AND PROVENANCE PROTOCOL

Each chunk is keyed by a stable identifier built from its source path and passage location. The persisted record stores: source ID, source case ID where available, citation, exact decision date, court, local PDF file, page number, character start and end, and chunk text. Retrieval runs record query ID, year, query text, index version, temporal policy, rank, score, and temporal status.

The renderer may expose only supplied selected evidence. Material content remains verbatim; authority cards and conclusions refer to evidence IDs. Verification then checks corpus existence, exact field equality, run membership, duplicate status, and temporal eligibility, with answer-key matching only after a displayed citation has passed. Parallel reporter forms are reconciled through stable source identity or normalised title plus exact decision date, never loose string similarity.

This protocol distinguishes four failure states that a single “correct citation” label would collapse:

- 1) a fabricated or altered citation;
- 2) a real citation that was not retrieved for this query;
- 3) a retrieved citation violating temporal or duplicate policy;
- 4) a fully valid citation that does not match the predefined expected authority.

In our results, states (1)–(3) occur zero times and state (4) occurs in 18 of 30 cases — a profile that a single aggregate score would render invisible.

VIII. RESULTS

A. Reporting populations

Four populations are reported and never aggregated across kinds: 1,503 eligible fixed ILDC test cases; a combined 37-case reference-evidence population (Base-30 frozen key with 150 citations, Extension-7 additive cases with 35, Combined-37 with 185), always stratified; and paired explanation-review cases (seven formative author-reviewed pairs, superseded by the 14-case LLM evaluation below).

B. Outcome prediction

On the 1,503-case population (Table II), sparse E1 reached accuracy 0.61344 and macro-F1 0.612342, exceeding corrected E2 mean-logit pooling at 0.596806 and 0.592358. E2’s secondary majority-vote aggregation reached 0.6015 and 0.5937. All primary models exceeded the 0.5017 majority-class accuracy.

This does not show domain pre-training is ineffective in general, only that the sparse baseline performed better under

TABLE II
OUTCOME PREDICTION. POPULATIONS ARE DISTINCT AND NOT A
LIKE-FOR-LIKE LEADERBOARD.

Model	n	Acc.	Macro-F1
Majority class	1,503	0.5017	0.3341
E1 TF-IDF + LR	1,503	0.61344	0.612342
E2 InLegalBERT, mean logits	1,503	0.596806	0.592358
E2 InLegalBERT, maj. vote	1,503	0.6015	0.5937
E3 evidence-augmented	30	0.666667	0.603175
E4 verified evid.-augmented	30	0.666667	0.603175

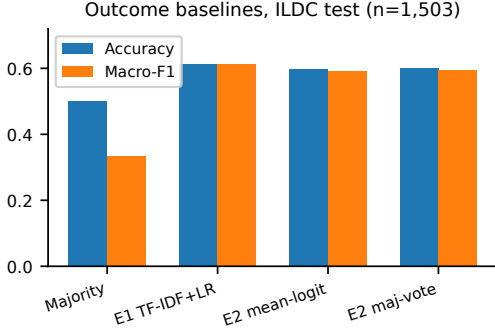


Fig. 1. Accuracy and macro-F1 for E1, corrected E2 under both pooling rules, and the majority baseline on the eligible ILDC test population ($n = 1,503$).

these frozen settings; Section IX shows the two models make substantially non-identical errors.

On the separate 30-case subset, evidence-augmented E3 and verified E4 both reached accuracy 0.666667 and macro-F1 0.603175, with identical confusion matrices. On the combined 37 cases the same shared predictor reaches accuracy 0.648649 and macro-F1 0.607347 with confusion matrix $[[6,3],[10,18]]$, again identical for E3 and E4 (Table III). **The equality is expected and is not evidence that verification is redundant:** with no evidence rejected, the predictor received an identical input; E4’s contribution is control, traceability, and explanation. Had verification rejected evidence, the reported pair would diverge visibly.

C. Authority recovery and evidence integrity

Base-30 recovery is Recall@5 0.40 (12/30) and Recall@100 0.50 (15/30: 12 selected + 3 retrieved-but-unselected); 15 authorities were absent at $k=100$.

The principal RQ1 result uses the combined 37-case population (Table III). The seven extension cases contribute no additional top-5 recovery (0/7) but five further top-100 recoveries at ranks 14, 16, 26, 32, and 89, with two absent; their displayed citations are likewise fully verified (35/35). The combined figures are therefore Recall@5 $12/37 = 0.324324$ and Recall@100 $20/37 = 0.540541$ over 185 displayed citations. The lower combined Recall@5 is descriptive of the added stratum, not evidence about the base configuration, and the frozen 30-case result above is retained unchanged alongside it.

TABLE III
RQ1 RETRIEVAL AND EVIDENCE-AUGMENTED PREDICTION BY STRATUM.
THE COMBINED ROW IS THE PRINCIPAL EVALUATION; STRATA ARE NEVER
POOLED SILENTLY.

Population	n	R@5	R@100	E3 Acc.	E3 F1
Base-30	30	0.40 (12/30)	0.50 (15/30)	0.6667	0.6032
Extension-7	7	0.00 (0/7)	0.7143 (5/7)	0.5714	0.5714
Combined-37	37	0.324324 (12/37)	0.540541 (20/37)	0.648649	0.607347

TABLE IV
EVIDENCE RECOVERY AND INTEGRITY ON THE FROZEN BASE-30
ANSWER-KEY STRATUM ($n = 30$ QUERIES; 150 DISPLAYED CITATIONS).
COMBINED-37 VALUES ARE REPORTED IN TABLE III AND THE
SURROUNDING TEXT.

Measure	Value	Arithmetic
Recall@5	0.40	12 of 30 in top 5
Recall@100	0.50	15 of 30 in top 100
Absent at $k=100$	0.50	15 of 30
Authority precision	0.08	12 of 150 shown
struct. ceiling	0.20	30 of 150 (5 shown, 1 credited)
Grounding	1.00	150/150 passed
Provenance validity	1.00	150/150 passed
Temporal violations	0	0 of 150
Unsupported claims	0	0 of 30 cases

Figure 2 shows the recovery funnel. Base-30 integrity is 150/150 grounded and provenance-valid with 0/150 temporal violations and 0/30 unsupported claims (Table IV, Fig. 3); the candidate log held 3,000/3,000 eligible candidates, and a five-case faithfulness check resolved every reference to its persisted chunk. Combined-37 holds the same profile at 185/185, 0/185, and 0/37.

E4 verification is fail-closed: any failing check rejects the citation. On live data nothing was rejected — a ceiling effect of already-valid selected evidence, not a reduction magnitude. Three positive controls through the unchanged verifier show the gate bites: altered passages rejected 37/37, fabricated citations 37/37, and backdated cases 37/37.

This is the paper’s central result. Verification success and authority recovery are dissociated: the system verified everything it displayed (185/185 combined; 150/150 on the frozen base) while recovering the expected authority in 20 of 37 cases at $k=100$ (15/30 base). Neither number alone describes the system. A reviewer told only that all displayed citations are valid would over-trust it; a reviewer told only that Recall@100 is 0.54 would miss that nothing displayed was fabricated, misattributed, or anachronistic.

D. Interpreting authority-consistency precision

Authority-consistency precision was 0.08, with recall 0.40 and F1 0.133 — to be read structurally, not as a fabrication rate: with one credited authority per case and five displayed, perfect inclusion caps the measure at $30/150 = 0.20$, so 0.08 is 40% of ceiling.

The 138 non-matching displayed citations are *not* independently annotated for substantive legal relevance and cannot

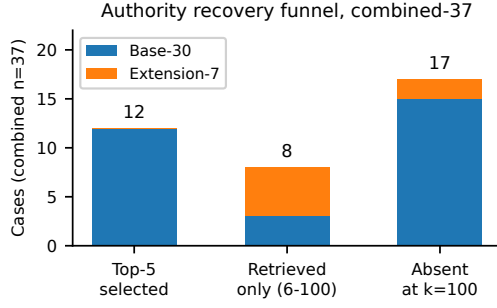


Fig. 2. Expected-authority recovery on the combined reference set ($n = 37$): 12 selected, eight retrieved but unselected (three base, five extension), 17 absent at $k=100$.

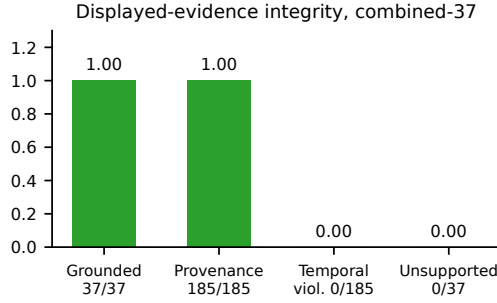


Fig. 3. Displayed-evidence integrity, combined $n = 37$ (185 citations): all grounding and provenance checks passed, with no temporal violations or unsupported claims. Base-30 contributes 150/150.

be declared incorrect. They passed every integrity check; they simply are not the one authority the key records. Reporting this metric without its ceiling would materially misrepresent the system.

E. Retrieval mechanism

The final configuration came from three bounded changes, each validated on the nine-case probe before the 30-case held-out test: full-input salient terms took the probe from 0/9 to 3/9 at $k=100$; the 80% coverage self-match condition took it to 6/9 and forced withdrawal of an over-broad lexical-mismatch claim; pre-ranking temporal filtering took it to 7/9 and, carried over, lifted Recall@5 from 5/30 to 12/30 and Recall@100 from 12/30 to 15/30 *without losing any of the 12 prior top-100 successes*.

Temporal capacity is not the whole story. 2013_35 stayed absent despite 79 eligible candidates in its raw top 100, while 1980_105 was recovered from only three and became a top-five hit: pool size and lexical relevance contribute independently, with residual lexical mismatch the prominent unresolved cause.

F. Explanation format

The principal RQ3 evaluation is a blinded exploratory presentation comparison by four LLM raters (Gemini, Claude

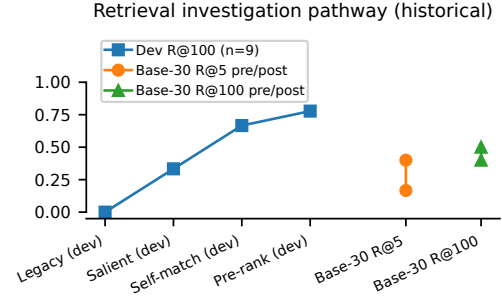


Fig. 4. Retrieval investigation pathway. The development probe ($n = 9$) covers salient-query and self-match repairs; the held-out comparison ($n = 30$) covers pre- to pre-ranking filtering. Distinct populations, not one learning curve.

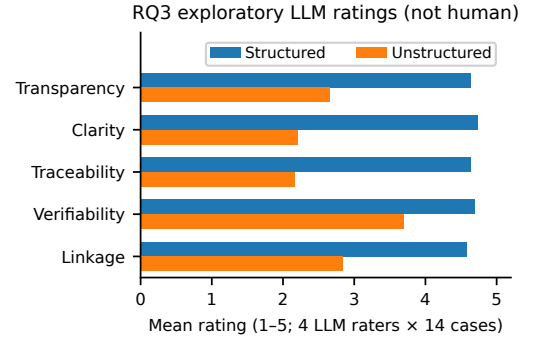


Fig. 5. Blinded LLM-rater means by dimension (14 cases $\times$ 4 raters): exploratory, not human-subject evidence.

Sonnet 4.6, DeepSeek, GPT-5.6 Luna) over 14 paired cases: the seven formative pairs plus all seven verified extension cases, with no cherry-picking. Each rater scored five transparency dimensions (1–5) with a forced preference, yielding 112 display ratings and 56 preferences. The structured presentation was preferred in 56/56 evaluations (+1.75 linkage, +1.00 verifiability, +2.48 traceability, +2.52 clarity, +1.98 transparency; overall +1.95, 4.66 vs. 2.71); see Fig. 5.

This is explicitly *not* human evaluation: no independent rater participated, no significance test was run, and nothing about human preference or legal correctness follows. Two cautions apply. First, DeepSeek and GPT-5.6 Luna produced byte-identical rating vectors, retained separately rather than merged, so the unanimity partly reflects duplicated outputs. Second, the seven-case self-review (means 4.57/2.71, 4.57/2.29, 4.43/2.86, 4.43/3.00; 7/7 structured) is formative, non-independent evidence superseded above. Its durable output remains negative: generic uncertainty wording did not adapt to 2013_35, suggesting generic uncertainty language may not adapt reliably to evidence quality.

TABLE V
OUTCOME-ERROR OVERLAP BY EVALUATION POPULATION. “E1” AND “E2” COUNT CASES WHERE ONLY THAT MODEL WAS CORRECT.

Population	Both	E1	E2	Neither
Full test ($n = 1,503$)	684	238	213	368
Answer key ($n = 30$)	18	3	3	6

TABLE VI
EXHAUSTIVE AUTHORITY-RECOVERY PARTITION ($n = 30$).

Outcome	n	Cases
Retrieved and selected	12	2008_1629, 1995_322, 1995_375, 1986_176, 1977_99, 1981_187, 1980_222, 1980_105, 1995_425, 2002_944, 1982_29, 1988_96
Retrieved, not selected	3	1980_133 (rank 15), 1981_55 (rank 28), 1985_40 (rank 78)
Absent at $k=100$	15	1997_792, 1993_185, 1971_295, 1974_36, 1986_378, 1984_136, 2013_35, 1980_217, 1978_33, 1977_145, 1995_412, 1995_403, 1986_397, 1994_632, 1992_84

IX. ERROR ANALYSIS

A. Outcome disagreement

The analysis is read-only over final frozen outputs; reconstructed E1 and inference-only E2 predictions reproduced their metrics before per-case joining. Across 1,503 cases, both models were correct on 684 and both wrong on 368; E1 alone was correct on 238 and E2 alone on 213 (Table V) — complementary errors suggesting ensemble or disagreement-triggered routing as future work.

In the evidence-augmented run, E2 was wrong while E3 and E4 were correct for two cases (1974_36, 1984_136). E3-correct/E4-wrong and E3-wrong/E4-correct each occurred in 0/30 cases, because verification changed neither the selected evidence nor the shared predictor input.

Critically, four cases (2008_1629, 1981_187, 1982_29, 1985_40) retrieved *and* selected the expected authority yet still produced an incorrect outcome prediction: correct recovery does not mechanically imply a correct prediction. No correct prediction was accompanied by an unsupported citation.

B. Authority-recovery buckets

Table VI gives the exhaustive partition: the middle bucket isolates *selection* failure (available but undisplayed, addressable by a better selector), while the absent bucket requires upstream improvement in representation, matching, ranking, or coverage.

C. Worked example: retrieval noise made visible

Case 1985_40 (factory at Atul; excise item 14D) shows why extractive presentation aids review. The five displayed

citations were verified, while the expected authority [1964] 2 S.C.R. 888 sat at rank 78 unselected.

Inspecting the displayed set, the first concerns excise valuation for Atul Products, but the others concern an industrial dispute, rayon and silk sales tax, and cigarette excise duty — mismatches the authority index and per-evidence blocks made visible, offered strictly as navigability illustration, not a relevance annotation.

X. LIMITATIONS AND THREATS TO VALIDITY

Scope. English-language Indian Supreme Court judgments only; no other Indian languages, High Court or lower-court material, separately versioned statutes, or cross-jurisdictional transfer.

Extraction approximation. OCR restored 12 of 15 affected PDF instances; three were excluded and repaired text may retain noise. With no gold facts/reasoning boundary, the facts rule can retain reasoning or omit facts near a boundary.

Answer-key size and era concentration. Evidence evaluation rests on 30 frozen cases, extended additively to 37 with seven later verified cases; 13 of the base 30 are from the 1980s. The combined set is neither large nor era-balanced and should not be read as representative of the ILDC test set or of Indian legal research generally — the primary constraint on generality.

Single reference authority. One expected authority per query, not every relevant source: inconsistency is a reproducible reference mismatch, not a substantive legal-relevance judgment.

Residual recovery gap. At $k=100$, 17 of the 37 expected authorities still went unrecovered; the 2013_35/1980_105 contrast shows temporal room alone does not surface the authority. This limited comparison does not establish a causal attribution for the remaining gap.

Year-level temporal granularity. ILDC query dates have year-level precision only, so any candidate sharing a query’s year is excluded rather than risk admitting a source that actually postdates it — a conservative choice that can discard genuinely eligible evidence. Day-level dates would require manually checking each case against a primary source; the limitation therefore sits in the dataset, not in how the pipeline parses dates.

Bundled verification. E4 checks several integrity properties together and cannot isolate any single component’s causal contribution.

Non-independent explanation review. The seven-case review is a non-random author self-review. The superseding 14-case LLM comparison is exploratory presentation evaluation by four model raters, not human-subject evidence; two runs produced identical outputs and are reported separately rather than merged, so even unanimity partly reflects duplicated runs.

Prototype scale. Results apply to the frozen implementation and samples rather than to all users, courts, domains, or changing corpora. No production deployment is claimed or evaluated.

XI. RESPONSIBLE USE AND GOVERNANCE

The system is designed as a legal-research aid, not an adjudicator or a provider of legal advice. E3/E4 predictions are experimental; the renderer states no legal conclusion beyond supplied evidence, predictions stay separate, and a human reviewer judges relevance, currency, authority, and applicability [3].

Risk controls are fail-closed: missing dates, same-year ambiguity, later dates, target identity, and duplication exclude evidence from display. A citation must resolve to exact corpus text and the correct retrieval run; altered or unsupported material is rejected. Stable locators make verification possible without trusting the renderer’s prose.

Governance also requires transparent negative results: discarded runs, superseded configurations, the collision correction, key replacements, OCR exclusions, and the non-independent review status are preserved. Populations and denominators remain separate, and alternative citations are not called irrelevant without annotation.

Public judgments do not remove privacy or misuse concerns: no public end-user search service is exposed and no consequential decision automated; any deployment would require access controls, licensing review, privacy assessment, independent evaluation, monitoring, and a correction process.

XII. CONCLUSION AND FUTURE WORK

We built and evaluated a provenance-preserving, temporally constrained evidence workflow for historical Indian Supreme Court research, keeping evidence and prediction measurements separate throughout.

The principal finding is a dissociation. Every one of 185 displayed citations across the combined 37 cases passed verification, with zero unsupported claims across the 37 cases, while 17 of the 37 expected authorities were absent at $k=100$. Verification succeeded for what the system displayed; it did not guarantee that the system found the expected authority. Legal-research systems therefore benefit from explicit controls — but those controls must be evaluated *alongside*, never *instead of*, retrieval coverage and human review.

The cross-corpus audit yields a transferable methodological lesson: only 11 of 5,391 syntactic identifier matches between ILDC and the eCourts collection were content-aligned. Identifier equality is a candidate hint requiring content validation, not evidence of judgment identity.

The supporting results are reported with their limits. On the 1,503-case outcome task the sparse TF-IDF baseline outperformed corrected InLegalBERT under the frozen settings — a result about this experiment, not a general verdict on legal-domain pre-training. Inference-time evidence augmentation reached 0.666667 accuracy and 0.603175 macro-F1 on the frozen 30-case subset (0.648649 and 0.607347 on the combined 37), descriptively and under acknowledged distribution shift. The 14-case structured-presentation comparison is exploratory LLM-rater evidence, not human preference; the seven-case self-review beneath it is formative and non-independent.

Future work. An era-balanced key of 75–100 cases would support a paired significance test (e.g., McNemar’s); a blinded review with independent raters and agreement reporting (e.g., Fleiss’ κ) would replace the formative explanation evidence; hybrid dense–sparse retrieval would test whether semantic retrieval closes residual lexical misses under the present controls; and evidence-sensitive uncertainty language would address the exposed calibration weakness.

REPRODUCIBILITY

All configurations, seeds, model revisions, corpus identities, and evaluation artifacts are frozen and machine-readable, including checkpoint-6318 (924a5bb9...dbdc773) with a stable replay hash. The implementation passes 81 automated research-pipeline tests (75 host-runnable plus 6 transformer/torch-only unit tests) and the documented reproducibility checks (14/14 documentation checks and a freeze audit of 39 recorded entries: 28 byte-exact, 1 after line-ending normalisation, and 10 differing only in non-scientific metadata, with all reported metrics verified identical; per-file evidence in the repository audit). Separately, the local demo stack is covered by 8/8 Spring Boot service tests, 6/6 FastAPI wrapper tests, and a passing TypeScript compile plus production frontend build; these exercise orchestration and contracts only, never the frozen scientific results. Code and artifacts are available at <https://github.com/0-YuvrajSingh/NyayaTrace>.

A minimal connected local research demo (React 18 + TypeScript UI, Spring Boot 3.2.5 orchestration API, FastAPI wrapper around the frozen pipeline) exposes the E4 evidence path with bearer-token authentication, per-request audit records, and experiment-metadata references; the FastAPI layer delegates every research step to the frozen implementation and the static frozen-artifact viewer is retained as a fallback. The stack binds to loopback only and is not a production, multi-tenant, or autonomous system.

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